Physical activity and insomnia in adolescents: the chain mediating role of anxiety and short video addiction

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Scientific Reports · 2026 · PMC13569844 · DOI: 10.1038/s41598-026-69547-7

Resumo (Abstract)

Adolescent insomnia constitutes a prominent public health priority, frequently co-occurring with psychological distress and digital media dependence. This study evaluated the distribution of shared variance among physical activity, anxiety, short video addiction and insomnia symptoms in adolescents. A cross-sectional survey was administered in December 2025 across five Chinese provinces, utilizing a convenience cluster sampling design to recruit 2,144 adolescents (14.8 ± 1.3 years). Structural path analysis was performed using the SPSS PROCESS macro to map the statistical trajectories among the target constructs while concurrently adjusting for key demographic covariates. Bivariate analysis revealed significant baseline correlations across all analyzed constructs. Path modeling indicated that physical activity maintained a significant negative association with insomnia scores ($\beta = -0.086$, $p < 0.001$). Total covariance analysis decomposed the data into a direct association accounting for 57.72% of the covariation matrix and three indirect pathways encompassing 42.28% of the shared variance. Specifically, the independent indirect pathways via anxiety and short video addiction accounted for 29.53% and 8.72% of the effect size, respectively, while the chain pathway captured a minimal but statistically significant 4.03%. Structural comparisons further confirmed that the hypothesized chain framework outperformed alternative causal configurations, with the baseline model yielding the lowest Akaike Information Criterion (42.35) and Bayesian Information Criterion (68.12). Rather than establishing definitive causal relationships, these cross-sectional findings indicate that elevated engagement in physical activity typically clusters with a reduction in both psychological and technological risk vectors, thereby supporting the integration of structured exercise into adolescent sleep screening frameworks.

Introduction

Insomnia is a prevalent clinical disorder characterized by persistent dissatisfaction with sleep quantity or quality. Its core clinical manifestations encompass difficulty initiating sleep (DIS), difficulty maintaining sleep (DMS), and early-morning awakening (EMA) 1 , all of which exert far-reaching detrimental impacts on individual somatic health, cognitive performance, and public safety. Patients suffering from chronic insomnia typically present with daytime fatigue, compromised emotional regulation, and pronounced cognitive deficits, particularly regarding attentional allocation and executive control 2 . Prolonged sleep deprivation has also been confirmed as the core risk factor for the onset of depression 3 . The prevailing theoretical framework in contemporary sleep medicine, the hyperarousal theory (HAT), posits that insomnia is not merely a passive regulatory deficit but rather stems from persistent, round-the-clock hyperactivity of the autonomic nervous, neuroendocrine, and cognitive-emotional systems 3 . At the physiological level, individuals with insomnia exhibit hypothalamic-pituitary-adrenal (HPA) axis dysfunction, characterized by abnormal nocturnal elevations in adrenocorticotropic hormone and cortisol secretion 2 , 4 . Neurobiologically, patients fail to down-regulate cortical regions associated with emotional processing and conscious awareness during the transition to sleep, thereby disrupting sleep onset 3 . This pathophysiological progression is systematically conceptualized by the “3P model”: predisposing factors, including genetics, gender,

psychological traits, establish the premorbid susceptibility of the individual 5 , 6 ; on this premise, precipitating factors (e.g., academic pressure and interpersonal conflicts) trigger the onset of acute insomnia 1 , 2 ; finally, perpetuating factors, primarily maladaptive compensatory behaviors and cognitive sleep misperceptions, drive the chronification of these acute disturbances into persistent sleep-wake cycle disorders 4 , 6 . Because adolescence represents a developmental window characterized by profound neural plasticity and psychosocial shifts, sleep disturbances during this stage can severely derail neurodevelopment, impair academic performance, and trigger emotional dysregulation 2 . Epidemiological surveys indicate that approximately 21% of Chinese adolescents experience moderate-to-severe insomnia symptoms 7 . Consequently, exploring non-pharmacological interventions to mitigate adolescent sleep disorders is of paramount public health importance, offering a scientific foundation for establishing an integrated “exercise-psychology-sleep” health promotion framework.

Physical activity (PA) is defined as any bodily movement produced by skeletal muscles that results in energy expenditure 8 . Accumulating epidemiological and clinical evidence demonstrates that regular, moderate-intensity PA exerts robust protective and therapeutic effects against insomnia. Long-term active cohorts report significantly lower rates of DIS, DMS, and EMA than their sedentary peers 9 . Clinical trials have further confirmed that initiating moderate-intensity exercise regimens significantly reduces Insomnia Severity Index scores, even among previously inactive individuals 10 . Additionally, PA contributes to the stabilization of sleep duration and the reduction of excessive daytime sleepiness 11 . Mechanistically, structured exercise modulates autonomic nervous system and cardiovascular regulation, thereby facilitating the optimization of sleep architecture 12 , 13 ; psychologically, PA serves as an adaptive coping strategy that dampens anxiety and depressive symptoms, cushioning the negative impact of daily stressors on sleep health 14 , 15 . Within contemporary Chinese socioeconomic context, adolescents face a dual burden of declining PA participation and escalating insomnia risk 14 . Constrained by intense academic competition, the PA levels of Chinese adolescents fall significantly short of the threshold recommended by the World Health Organization (WHO) 16 . This low-activity, high-stress lifestyle not only induces sleep disturbances, but also clusters with broader mental health vulnerabilities 15 . Accordingly, evaluating the efficacy of PA as a scalable, non-pharmacological intervention to alleviate insomnia represents an urgent clinical priority 10 . This line of inquiry aims to protect adolescent developmental trajectories and buffer academic stress. Based on these observations, hypothesis H1 is proposed: PA negatively predicts insomnia.

Anxiety is a complex psychophysiological state characterized by cognitive hyperarousal—manifesting as excessive apprehension regarding future uncertainties—and somatic tension, such as accelerated heart rate and muscular hypertonicity 17 . Longitudinal and cross-sectional inquiries have consistently established a positive correlation between anxiety severity and insomnia indicators, whereby highly anxious individuals exhibit a elevated risk of DIS and DMS 18 , 19 . Baseline anxiety has been identified as an robust clinical predictor of subsequent chronic insomnia 20 . This pathological link operates across cognitive and physiological domains: cognitively, pre-sleep cognitive rumination concerning daytime stressors directly interferes with the quiet state necessary for sleep initiation 17 , 21 ; physiologically, anxiety perpetuates central and autonomic nervous system hyperarousal, precluding the somatic relaxation required for sleep transition. Concurrently, elevated anxiety co-occurs with alternations in sleep architecture, demonstrating synchronized shifts with circadian disruptions and neuroendocrine profiles 22 . Influenced by demanding academic workloads and family expectations, anxiety is highly prevalent among Chinese adolescents, making them particularly vulnerable to sleep disturbances 23 . Given that adolescent sleep deficits not only impair immediate

cognitive functioning 18 but also serve as a precursor to subsequent depressive episodes 20 , clarifying the mechanisms linking anxiety and insomnia carries significant clinical weight. Preventing this vicious cycle is vital for safeguarding adolescent psychological well-being. Thus, hypothesis H2 is proposed: anxiety mediates the relationship between PA and insomnia.

Within contemporary digital landscapes, short video addiction (SVA) represents a prominent correlate of sleep disturbances, exhibiting cross-sectional inverse associations with physical activity duration and positive associations with elevated physiological arousal scores. Characterized by a compulsive psychological dependence and an inability to regulate use, SVA represents an extension of behavioral addiction paradigms into the short-form mobile media domain 24 . Unlike traditional internet use, short-form video applications provide highly customized, algorithm-driven feeds and high-frequency sensory stimuli, inducing rapid and potent psychological gratification 25 . Empirical studies suggest that SVA is associated with elevated insomnia severity. On one hand, excessive screen time leads to media-induced sleep displacement, directly shortening total sleep duration 26 ; on the other, the high cognitive and emotional engagement triggered by personalized content increases pre-sleep arousal, while the blue light emitted by mobile devices suppresses melatonin secretion and delays circadian phases 25 , 27 . Consequently, SVA has become a critical perpetuating factor in adolescent sleep disturbances 24 , 26 . Recent large-scale surveys indicate that approximately 31.06% of adolescents report sleep-related complaints, a proportion that is significantly higher among individuals with severe SVA 28 . Investigating this behavioral vector is highly relevant for understanding the multifaceted etiology of modern sleep disorders. Hypothesis H3 is therefore proposed: SVA mediates the relationship between PA and insomnia.

Current research suggests that anxiety and SVA are not isolated phenomena but rather interact dynamically. As a maladaptive coping mechanism, SVA often functions as an immediate emotional escape for individuals experiencing psychological distress 29 , 30 . According to the Conservation of Resources Theory (CRT), when adolescents deplete their cognitive and emotional resources due to academic demands, they frequently turn to passive media consumption to relieve immediate anxious tension 30 . However, this compensatory measure will cause further resource depletion 29 , leading to bedtime procrastination, trapping individuals in a maladaptive “anxiety-overuse-intensified anxiety” cycle. Statistically, anxiety is also linked to elevated perceptual and cognitive loads, diminished indicators of attentional allocation, and a higher empirical probability of engaging with fragmented, instant-feedback short video platforms. This reinforces addictive tendencies, which in turn encroach on sleep time, amplify anxious arousal, disrupt integrity of sleep structure, and escalate pre-sleep physiological arousal 31 . Indeed, empirical data indicate that over two-thirds of frequent short video users experience post-use sleep disturbances, which are strongly tied to heightened anxiety 32 . This process of evolution from psychological state to behavioral deviation may constitute a critical path for PA to affect insomnia. Accordingly, this study proposes hypothesis H4: anxiety and SVA play a chain mediating role in the relationship between PA and insomnia.

The hypothesized sequencing of variables in this study is grounded in a structural taxonomy of psychological phenomena, moving from a relatively stable behavioral trait to a fluctuating cognitive state, and ultimately manifesting as a compulsive behavioral output. Under the auspices of mood management theory, core psychological distress and affective vulnerability typically precede maladaptive technology-seeking behaviors, as individuals tend to leverage media algorithms characterized by instantaneous gratification for emotional regulation 33 . This baseline ordering establishes anxiety as an explanatory antecedent to short video dependence, rather than merely a symptomatic byproduct. Although alternative directional vectors cannot be statistically eliminated within a cross-sectional matrix, our chosen sequence aligns with standard resource-depletion

frameworks: physical movement preserves cognitive control resources, lower anxiety protects baseline regulatory capacity, and attenuated digital immersion preserves the psychological preconditions necessary for sleep induction.

To summarize, while the protective effects of PA on sleep have been documented, the interactive, sequential roles of anxiety and SVA within this regulatory pathway remain poorly understood. By constructing a chain mediation model (Fig. 1), this study aims to clarify the pathways through which exercise habituation may alleviate adolescent insomnia, providing empirical insights for targeted clinical and educational interventions.

Fig. 1 Chain mediation model.

Participants and methods

Participants

This study officially launched in December 2025. Following the feasibility principle, the research team relied on existing inter-institutional cooperation and teacher contact network, and selected Hunan, Sichuan, Shaanxi, Shanxi and Anhui as survey sites through assistance of relevant managers. The sample spanned 7 distinct secondary schools, comprising one 12-year comprehensive school (encompassing junior and senior high divisions) in Hunan Province, three junior high schools in Sichuan Province, one junior high school in Shaanxi Province, one senior high school in Shanxi Province, and one senior high school in Anhui Province. A cluster convenience sampling method was adopted, with survey among students of the selected schools by class. Participants were included in the study if they met the following criteria: possessing sufficient cognitive and comprehension capacities to complete the survey independently, and providing explicit informed consent prior to participation (with parental or legal guardian consent obtained for all minors). This investigation used paper questionnaires with an informed consent form attached to the first page. It provided a detailed explanation of the research purpose, data anonymity, confidentiality and voluntary participation principles, and clarified participants' right to withdraw unconditionally at any time. The survey procedure proceeded as follows: summoned the head teachers of selected classes to hold a briefing session; research administrators then entered the class to explain the study, distribute questionnaires, clarify response instructions and supervise group completion. Completed questionnaires were collected on site. A total amount of 2282 questionnaires were returned. A rigorous screening protocol was applied, whereby questionnaires were excluded from the final analytical sample if they met any of the following criteria: contained missing values or omitted items; displayed multiple selections for single-choice items; exhibited patterned responses; or lacked completed informed consent procedures. The final analytical sample comprised 2144 participants (14.8 ± 1.3), yielding an effective response rate of 93.95%, consisting of 1075 males (50.1%) and 1069 females (49.9%). Among these, 810 (37.8%) held urban household registration and 1334 (62.2%) held rural household registration. Detailed demographic characteristics are presented in Table 1 .

Table 1 Demographic profile of participants. Features Categories N Percentage(%) Gender Male 1075 50.1 Female 1069 49.9 Only-child status Yes 146 6.8 No 1998 93.2 Residence Urban 810 37.8 Rural 1334 62.2 Boarding status Yes 1753 81.8 No 391 18.2

Instruments

PA

PA among adolescents was evaluated using the Physical Activity Rating Scale (PARS-3) developed by Liang et al. 34 . By systematically examining 3 dimensions—intensity, duration, and frequency of exercise—this instrument provides quantitative data to objectively appraise individual engagement in physical activities over the preceding month, demonstrating stable test-retest reliability and high criterion validity within adolescent cohorts 35 . The scale comprises 3 items rated on a 5-point scale. The total physical activity volume was operationalized using the following formula: PA volume = intensity × (duration-1) × frequency. Higher cumulative scores indicate a greater volume of physical exercise.

(2) Anxiety

The Generalized Anxiety Disorder-2 (GAD-2), developed by Spitzer et al. was employed to evaluate anxiety levels among adolescents 36 . Serving as a brief, rapid psychometric screening tool, this instrument is designed to capture self-reported symptom severity and risk stratification within a short time window, possessing robust internal consistency and construct validity 37 , 38 .The scale contains 2 items rated on a 4-point scale (0 = not at all to 3 = nearly every day), higher scores denote greater vulnerability to anxiety. In the present study, this scale yields a Cronbach's α coefficient of 0.80.

(3) SVA

SVA in adolescents was measured using the Problematic Short Video Media Use Scale (PSVMUS) constructed by Mao et al. 39 . This instrument encompasses three dimensions: behavioral-cognitive shifts, physiological discomfort, and social viscosity. It has been extensively validated within adolescent samples, displaying a highly stable factor structure 40 , 41 . The instrument consists of 13 items scored on a 5-point scale (1 = strongly disagree to 5 = strongly agree). Higher summed scores indicate more severe dependence on short video platforms. In the current sample, the Cronbach's α coefficient for this scale is 0.86.

(4) Insomnia

Insomnia severity were assessed using the Athens Insomnia scale (AIS) developed by Soldatos et al. 42 . Designed based on the diagnostic criteria of the International Classification of Diseases (ICD-10), this instrument is characterized by high applicability and reliability, demonstrating robust diagnostic validity in screening sleep disorders across clinical and epidemiological populations 43 , 44 . The scale includes 8 items rated on a 4-point scale (0 = none to 3 = severe), where higher scores denote more pronounced insomnia symptoms. In this study, the Cronbach's α coefficient for AIS is 0.86.

Covariate control

To accurately estimate the predictive effect of PA on insomnia and isolate confounding influences from irrelevant variables, this study incorporated the following demographic indicators as covariates for statistical control:

Gender and age were controlled to account for biological differences and physical and psychological development that may shape sleep patterns. Grade and boarding status were included to adjust for disparities in academic pressure and the impact of residential arrangements on daily routines. Residence, only-child status, and socioeconomic status were controlled to reduce heterogeneity in developmental contexts and family structures, which may affect quality of life and mental health.

By adjusting for these covariates, the analysis was conducted under a statistically cleaner framework, allowing clearer examination of the complex pathways through which PA influences insomnia in

adolescents via anxiety and SVA. This approach improves the precision of causal inference and strengthens the robustness and generalizability of the findings.

Data analysis

Data processing was performed using SPSS 25.0. To enhance the interpretability of interaction terms and cross-model comparability, all continuous variables were standardized prior to model entry; thus, all coefficients reported herein are standardized. Common method bias was examined through factor analysis. Pearson correlation was then applied to assess the strength of associations among study variables. The significance of indirect and conditional effects was evaluated using a percentile Bootstrap resampling procedure with 5,000 iterations. An effect was deemed statistically significant if its 95% bias-corrected confidence interval (95% CI) excluded zero. Diagnostic assessments were conducted prior to mediation analysis to examine multicollinearity, residual distribution, heteroscedasticity, and influential observations. The variance inflation factor ranged from 1.00 to 1.99 with tolerance values between 0.50 and 1.00, indicating the absence of severe multicollinearity. The maximum Cook's distance was 0.020, suggesting that no single observation exerted excessive influence on the estimates. Residual skewness ranged from – 0.06 to 0.91, and excess kurtosis ranged from 0.05 to 1.74. Given that the Breusch-Pagan test indicated mild heteroscedasticity in the anxiety and insomnia equations, sensitivity tests were performed utilizing HC3 heteroscedasticity-robust standard errors, which confirmed that the direction and statistical significance of the core pathways remained unaltered.

Results

Common method bias test

Results from the Harman single-factor test showed that 24 factors had eigenvalues greater than 1, with the first factor accounting for 28.40% of the total variance. This proportion fell below the 40% threshold, indicating no severe common method bias in this study.

Correlation analysis

As shown in Table 2 , statistically significant correlations emerged among all 4 variables.

PA was negatively associated with anxiety, SVA, and insomnia (r = – 0.108, r = – 0.103, r = – 0.162), providing preliminary support for the suppressive effect of regular PA on psychological and behavioral disturbances. Anxiety was positively linked to SVA and insomnia (r = 0.318, r = 0.510), while SVA was likewise positively correlated with insomnia (r = 0.353). These patterns point to strong coherence across risk indicators in the psychological and behavioral domains.

Table 2 Correlation matrix of variables. Variables M ± SD PA Anxiety SVA Insomnia PA 15.20 ± 17.87 - Anxiety 3.89 ± 1.69 – 0.108** - SVA 37.55 ± 10.60 – 0.103** 0.318** - Insomnia 15.63 ± 4.78 – 0.162** 0.510** 0.353** - **: $p < 0.01$.

Structural model comparison and alternative path testing

Given the cross-sectional nature of the data, the structural directional validity of the hypothesized model (PA → Anxiety → SVA → Insomnia) was empirically evaluated against three alternative configurations. Because these modifications yielded non-nested structural designs, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) were employed as primary benchmarks for statistical model selection, supplemented by traditional fit indices. Lower AIC and BIC

values reflect a more parsimonious model that achieves a superior reproduction of the empirical variance-covariance matrix.

Grounded in plausible alternative causal arrangements, three competing frameworks were specified: Competing Model 1 positioned insomnia as the upstream predictor of the entire pathway; Competing Model 2 inverted the sequential order of the mediators and insomnia; and Competing Model 3 specified a non-recursive structure incorporating bidirectional paths between anxiety and insomnia.

As presented in Table 3 , the hypothesized model yielded the most ideal fit indices (χ^2 / df = 1.84, CFI = 0.985, TLI = 0.979, RMSEA = 0.024, SRMR = 0.021). Concurrently, compared to all competing configurations, the hypothesized pathway sequence generated the lowest AIC (42.35) and BIC (68.12), whereas the alternative models exhibited noticeable degradation in fit indices (Competing model 1: AIC = 115.42, BIC = 141.19). These statistical parameters provide robust empirical support for the current chain mediation layout, demonstrating that the hypothesized path configuration represents the most parsimonious and data-supported representation of the observed variables.

Table 3 Comparison of fit indices between the baseline model and competing models. Model configuration χ^2 / df CFI TLI RMSEA SRMR AIC BIC Baseline Model PA → Anxiety → SVA → Insomnia 1.84 0.985 0.979 0.024 0.021 42.35 68.12 Competing Model 1 Insomnia → Anxiety → SVA → PA 4.62 0.912 0.895 0.068 0.074 115.42 141.19 Competing Model 2 PA → Insomnia → SVA → Anxiety 3.98 0.925 0.903 0.059 0.065 98.74 124.51 Competing Model 3 PA → Anxiety → Insomnia → SVA 2.15 0.971 0.962 0.035 0.042 56.18 89.45

Chain mediation effect test

After accounting for potential confounding effects of gender, age, residence and other demographic factors, mediation effects were tested in this study. As presented in Table 4 , PA negatively predicted anxiety (β = – 0.101, t = – 4.624). Anxiety was positively associated with SVA (β = 0.308, t = 14.971), and PA negatively predicted SVA (β = – 0.063, t = – 2.990). SVA positively predicted insomnia (β = 0.205, t = 10.790), as did anxiety (β = 0.433, t = 22.781). PA further exerted a direct negative predictive effect on insomnia (β = – 0.086, t = – 4.685). Bootstrap analyzes revealed that none of the 95% confidence intervals for the indirect pathways included 0, confirming statistical significance for all indirect effects in the model.

Table 4 Regression analysis of variable relationships. Dependent variables Independent variables R R^2 F β t p 95% CI LB UB Anxiety 0.129 0.017 4.521 PA – 0.101 – 4.624 0.000 – 0.144 – 0.058 Gender 0.021 0.475 0.635 – 0.066 0.107 Age 0.017 0.754 0.451 – 0.027 0.060 Grade 0.054 1.057 0.291 – 0.046 0.155 Residence – 0.011 – 0.222 0.824 – 0.106 0.085 Only-child status 0.152 1.766 0.078 – 0.017 0.322 Boarding status – 0.105 – 1.730 0.084 – 0.224 0.014 Socioeconomic status 0.004 0.365 0.715 – 0.019 0.027 SVA 0.331 0.109 29.137 PA – 0.063 – 2.990 0.003 – 0.104 – 0.022 Anxiety 0.308 14.971 0.000 0.268 0.349 Gender 0.088 2.087 0.037 0.005 0.170 Age 0.011 0.541 0.588 – 0.030 0.052 Grade – 0.020 – 0.417 0.677 – 0.116 0.076 Residene 0.012 0.255 0.799 – 0.079 0.103 Only-child Status 0.095 1.158 0.247 – 0.066 0.256 Boarding status 0.001 0.012 0.990 – 0.113 0.114 Socioeconomic status 0.019 1.676 0.094 – 0.003 0.040 Insomnia 0.561 0.315 97.896 PA – 0.086 – 4.685 0.000 – 0.122 – 0.050 Anxiety 0.433 22.781 0.000 0.396 0.470 SVA 0.205 10.790 0.000 0.168 0.242 Gender 0.063 1.721 0.085 – 0.009 0.136 Age 0.008 0.407 0.684 – 0.029 0.044 Grade 0.038 0.893 0.372 – 0.046 0.123 Residence – 0.097 – 2.379 0.017 – 0.177 – 0.017 Only-child status – 0.070 – 0.976 0.329 – 0.212 0.071 Boarding status – 0.072 – 1.427 0.153 – 0.172 0.027 Socioeconomic status – 0.005 – 0.557 0.577 – 0.025 0.014

Table 5 presents the decomposition of effect sizes, mapping the statistical variance shared between physical activity and insomnia scores through the designated pathways.

The direct of PA on adolescent insomnia accounted for the largest share of the total effect, at 57.72%. Bootstrap testing indicated that the total indirect effect of PA on insomnia via the mediating pathways was statistically significant. The absence of zero within the 95% bias-corrected percentile confidence interval confirmed the presence of a significant overall mediating mechanism, with the total indirect effect accounting for 42.28% of the total effect. The mediating role of anxiety was the most prominent, contributing 29.53% of the total effect. The mediating effect of SVA followed, at 8.72%, while the chain mediation pathway through anxiety to SVA accounted for 4.03%.

These findings provide consistent support for the theoretical model proposed in this study. Anxiety and SVA not only function as independent mediators between PA and insomnia in adolescents, but jointly form a sequential mediating pathways linking PA to sleep disorders (Fig. 2).

Table 5 Path and effect decomposition of the chain mediation model. Effect Path Effect size Proportion of total effect SE 95% CI LB UB Direct effect PA → Insomnia – 0.086 57.72% 0.018 – 0.122 – 0.050 Total indirect effect PA → Insomnia – 0.063 42.28% 0.011 – 0.085 – 0.042 Mediation effect PA → Anxiety → Insomnia – 0.044 29.53% 0.009 – 0.063 – 0.027 Mediation effect PA → SVA → Insomnia – 0.013 8.72% 0.004 – 0.022 – 0.005 Mediation effect PA → Anxiety → SVA → Insomnia – 0.006 4.03% 0.002 – 0.010 – 0.004 Total effect PA → Insomnia – 0.149 100% 0.022 – 0.192 – 0.107 Note: N = 2144. Effect sizes reported in the table represent standardized coefficients estimated via path analysis. SE = standard error; CI = confidence interval; LB = lower bound; UB = upper bound.

Fig. 2 Mediation model of anxiety and SVA between PA and Insomnia.

Discussion

The empirical findings of this study demonstrate that PA significantly and negatively predicts adolescent insomnia, thereby validating Hypothesis H1. This outcome aligns with a robust body of literature detailing the sleep-promoting benefits of regular exercise 45 . Pathway analysis reveals that the direct effect of PA on insomnia accounts for 57.72% of the total effect, proving that habitual exercise serves as a powerful, non-pharmacological strategy to directly mitigate sleep disturbances 46 . The statistical covariation between higher PA and lower insomnia indicators reflects an empirical link embedded with interconnected physiological and psychological markers. Physiologically, this relationship is frequently elucidated through the “thermogenic hypothesis”. Engaging in exercise induces a transient elevation in core body temperature. During the subsequent post-exercise recovery period, a compensatory rapid downregulation of core body temperature occurs. This thermal fluctuation is hypothesized to stimulate heat-sensitive neurons within the preoptic area of the anterior hypothalamus, thereby facilitating sleep onset and enhancing deep, slow-wave sleep duration 45 . Concurrently, regular physical exercise serves as a potent zeitgeber that helps calibrate circadian rhythms 47 , By increasing daytime energy expenditure and light exposure, PA assists in aligning the biological clock, which ultimately alleviates DIS and DMS 48 . Beyond these somatic pathways, PA attenuates insomnia through psychological channels. Chronic sleep disturbances are deeply intertwined with maladaptive cognitive patterns and affective distress; in this context, habitual exercise enhances general self-efficacy and serves as an emotional buffer against daily stress. Optimizing this psychological profile correlates with lower variance in nocturnal cognitive rumination and hyperarousal indicators among individuals reporting sleep disturbances, helping them enter and maintain deep sleep 46 , 49 . Given that adolescence represents a critical developmental window

marked by intensifying academic pressure and escalating vulnerability to sleep disorders 50 , identifying low-cost, accessible interventions is of paramount importance. As an easily implementable lifestyle component, structured PA programs not only directly cushion adolescent sleep quality against developmental stressors 51 but also lay a sustainable foundation for long-term psychological and physical well-being 52 .

The results show that anxiety plays a mediating role between PA and insomnia, supporting hypothesis H2 and complementing established psychophysiological frameworks 53 , 54 . The detrimental impact of anxiety on sleep architecture is multifaceted, operating across integrated cognitive and neuroendocrine domains 55 . At the physiological level, chronic anxiety-induced distress triggers persistent hyperactivity of the HPA axis, leading to abnormal nocturnal elevations of corticotropin-releasing hormone and cortisol. This sustained hormonal surge maintains the central nervous system in a state of hyperarousal, directly impeding sleep initiation and promoting sleep fragmentation. Cognitively, the excessive apprehension and somatic tension (e.g., muscular hypertonicity, cardiovascular reactivity) typical of anxious states fuel pre-sleep cognitive rumination. This heightened vigilance over sleep performance itself initiates a reinforcing loop, where the dread of sleeplessness perpetuates somatic arousal, further exacerbating DIS and DMS 56 . Anxiety also triggers the individuals' attention and alert system, chest tightness and other physical symptoms, making the brain unable to enter the required desynchronization state, resulting in increased sleep fragmentation 57 . Crucially, regular PA serves as an effective behavioral buffer that disrupts this pathological cycle 58 . Neurobiologically, consistent exercise optimizes the body's allostatic mechanisms. By promoting adaptive plasticity in the neuroendocrine system, exercise enhances glucocorticoid receptor sensitivity, thereby lowering HPA axis reactivity and dampening systemic hyperarousal 45 , 57 . Concurrently, PA stimulates the expression of brain-derived neurotrophic factor and modulates serotonergic pathways, which directly alleviates affective distress while stabilizing sleep architecture 59 . Psychologically, exercise provides adolescents with a vital cognitive diversion and an adaptive coping space. Any mechanistic interpretation regarding the efficacy of physical activity must be strictly framed within the mathematical boundaries of the study design. The observation that favorable psychological profiles and diminished compulsive media dependence cluster with active exercise habits 19 provides merely a non-directional baseline record. This evidence showcases a beneficial behavioral configuration wherein the severity of risk factors is statistically isolated, rather than proving a chronological, preventive sequence of causal nature. Because biomarker tracking and neuropsychological experimental tasks were beyond the scope of this large-scale survey, these potential pathways within the current sample must be treated strictly as speculative theoretical frameworks rather than empirically observed mechanisms. These interconnected paths represent a cross-sectional matrix description of the shared variance among behavioral indicators, highlighting a multidimensional lifestyle profile associated with lower insomnia scores without implementing direct causal trajectories.

Results of the study demonstrate that SVA mediates the relationship between PA and insomnia, validating hypothesis H3 and aligning with prior empirical investigations into the association between SVA and insomnia 26 , 60 . Owing to algorithm-driven reward systems and highly stimulating audiovisual content, the maladaptive use of short video applications exerts a powerful, multi-channel grip on adolescent sleep hygiene 61 . SVA is closely tied to the self-regulatory failure, compelling individuals to delay bedtime even without external interference 26 , 62 . This procrastination tendency gradually evolves into an avoidant coping mechanism: under the weight of academic stress or sleep-related anxiety, adolescents instrumentally seek the instant gratification offered by short video platforms, trapping themselves in chronic sleep deprivation 63 . Furthermore, excessive media

immersion also weakens executive function, especially inhibitory control 64 . This neurocognitive degradation hinders adolescents' capacity to disengage from screen-viewing before sleep, generating a negative feedback loop where bedtime frustration and self-criticism further fuel physiological arousal and exacerbate insomnia 65 . Within this framework, PA emerges as an effective behavioral intervention. From a cognitive perspective, regular exercise enhances executive functioning and strengthens self-regulatory capacity, enabling adolescents to more effectively suppress the compulsive urge to scroll through personalized algorithms 64 . This behavioral control directly curtails late-night viewing and prevents sleep displacement. Psychologically, the self-efficacy gains associated with sports participation can mitigate social anxiety, reducing the motivation to escape into virtual spaces or seek digital social compensation 66 . By restoring cognitive resources and reducing bedtime procrastination, PA supports the maintenance of regular, structured sleep-wake routines 26 , 63 .

Research results support that anxiety and SVA play a chain mediating role between PA and insomnia, confirming hypothesis H4. This finding indicates that the impacts of psychological distress and digital addiction are not merely parallel, independent vectors but are dynamically coupled within adolescent lifestyles. Confronted with intense academic competition and social evaluative pressure, contemporary adolescents frequently experience pervasive anxiety. Short video platforms, characterized by ultra-short feedback loops and highly personalized content, offer a convenient but maladaptive tool for emotional escape 67 , 68 . However, the intense cognitive and emotional engagement induced by these platforms elevates pre-sleep arousal, while device-emitted blue light disrupts melatonin secretion and circadian stability, ultimately resulting in severe sleep deficits 29 . As a multidimensional intervention, PA directly addresses both links of this chain. Moderate-intensity exercise stimulates the release of beta-endorphins and dopamine, providing positive affective feedback that lowers trait anxiety and stress reactivity 69 , 70 . Beyond emotional regulation, structured sports participation offers a natural physical substitution effect, occupying temporal windows that might otherwise be dedicated to passive screen-viewing, while reinforcing the self-regulatory control necessary to resist late-night digital temptations 71 , 72 . Beyond these potential psychological and behavioral pathways, PA is frequently associated with enhanced sleep, which may be linked to the stabilization of circadian alignment. For individuals heavily exposed to virtual social interactions and digital stimulation, the real-world sensory engagement inherent in physical exercise is often accompanied by a higher probability of maintaining regulated biological rhythms 73 . The structural path "PA $\rightarrow$ Anxiety $\rightarrow$ SVA $\rightarrow$ Insomnia" identifies a pattern of cumulative statistical variance; this orientation indicates that, within the cross-sectional distribution, variations in individual sleep indicators are a function of overlapping psychological scores and behavioral parameters. What the model captures is a covariation clustering effect, rather than establishing an active intervention capable of breaking or arresting behavioral cycles. Specifically, high levels of physical activity systematically covary with lower anxiety indices and attenuated digital media dependence. This non-directional feature aligns with prior reports characterizing physical exercise and reduced psychological and technological risks as concomitant states 69 , 72 . Although these cross-sectional data demonstrate the statistical compatibility of the hypothesized sequence, the current analytical matrix cannot rule out alternative directional vector paths.

It must be emphasized that although algorithm-driven reward mechanisms within short videos are theoretically mapped onto putative overactivation of dopaminergic pathways and subsequent circadian shifts, this investigation contains no biological validation of these trajectories. The observed statistical overlap between SVA scale scores and insomnia indicators is consistent with the hypothesis of "digitally induced neural arousal and biological clock disruption," yet it does not

constitute evidence of altered neurotransmitter signaling. Consequently, these mechanistic discussion points serve merely as potential, literature-derived theoretical explanations. They demonstrate the logical compatibility between the structural paths of the model and neurobiological theories, rather than providing a definitive empirical depiction of the neurochemical profiles of adolescents.

A critical limitation of this research framework rests on the fact that competing structural paths are mathematically equivalent within a cross-sectional distribution. Given the cross-sectional design, the directional causality of the pathway wherein physical activity influences insomnia via the mediation of anxiety and SVA warrants careful interpretation. The relationship between internal affective states and sleep disturbances is inherently non-linear and may exhibit reciprocal causality over long-term longitudinal observations. Extant literature suggests that chronic somatic and sleep-related distress frequently act as upstream triggers driving subsequent emotional dysregulation and generalized anxiety 74 , that is, the notions that primary sleep debt amplifies next-day affective instability, or that late-night insomnia prompts individuals to instrumentally seek highly stimulating digital content, remain equally plausible in theory. To rule out potential equivalent models and address alternative causal explanations, this study empirically tested several competing structural configurations. The results demonstrated that the hypothesized baseline model yielded superior fit and lower informational discrepancy compared to configurations positioning insomnia or SVA as upstream predictors. The marked degradation of fit indices in alternative models strongly reinforces that the “PA $\rightarrow$ Anxiety $\rightarrow$ SVA $\rightarrow$ Insomnia” pathway represents the most parsimonious and data-supported representation of the observed variance-covariance matrix, rather than a statistical artifact induced by bidirectional psychological distress. Concurrently, although the unidirectional empirical model demonstrated robust fit standards within the current sample, it must be interpreted as a prospective cross-sectional snapshot: the data herein do not refute competing bidirectional architectures but rather highlight a specific operative pathway wherein physical activity clusters with emotional stability, indirectly corresponding to diminished digital reliance and improved sleep indicators. Future longitudinal investigations are required to fully untangle these prospective dynamics.

In this study, the verification of the chain mediation effect not only clarifies the logical path of “PA $\rightarrow$ Anxiety $\rightarrow$ SVA $\rightarrow$ Insomnia”, but also deeply reveals the physical and mental adjustment mechanism of adolescents in the digital survival environment. Crucially, although this chain pathway achieved empirical significance within the structural model, its baseline effect size remains numerically modest. This limited statistical contribution implies that the chain evolution represents a micro-behavioral configuration rather than a dominant clinical driver of adolescent sleep quality. Consequently, from a practical framework, expecting to utilize physical activity as a standalone modality to arrest the “psycho-technological” cycle and thereby generate large-scale, immediate clinical reversals of insomnia is unrealistic. Mapping this pathway is practically valuable for identifying a subtle, complementary avenue for lifestyle screenings—specifically, discerning how exercise habits cluster with attenuated psychological and digital risk vectors. Rather than framing physical activity as a grand, independent therapeutic intervention, these data position it as an auxiliary, low-cost baseline component to be integrated into broader, multidimensional adolescent sleep hygiene programs.

Research limitations and prospects

Based on a large scale survey, this study established and verified a chain mediation model linking PA and insomnia in adolescents. Despite these contributions, several limitations remain due to practical research constraints.

The convenience cluster sampling strategy constrains the external validity of the model. Because participating students naturally cluster within specific schools and classrooms, their behavioral and sleep patterns may be shaped by shared environmental factors, such as institutional schedules and academic workloads. Treating the database as a single homogeneous whole, this study failed to mathematically control for these nested variations. Consequently, future research should adopt stratified random sampling designs and introduce advanced analytical frameworks capable of disentangling individual behavioral habits from broader institutional and environmental influences. Because all indicators were collected synchronously via uniform self-report instruments during a single assessment, the data matrix is structurally vulnerable to common method bias. Although diagnostic analysis included a post-hoc Harman one-factor extraction yielding a value below the standard 40% threshold, this approach lacks sufficient statistical power to isolate subtle systematic variance inflation or social desirability bias. While logistical constraints in large-scale screenings precluded the introduction of advanced structural controls, this psychometric limitation necessitates a conservative interpretation of effect sizes. Future inquiries must integrate objective sleep-tracking parameters to explicitly separate behavioral variance from method-induced covariation. While the chain mediation effect reached statistical significance, its overall effect size was relatively modest. This implies that other unmeasured confounding factors contribute to the development of adolescent insomnia. Future investigations may explore subgroup variations across grades and residential contexts, and examine targeted regulatory strategies addressing distinct anxiety sources. A notable methodological constraint of this research centers on the operationalization of the primary predictor. Although the single-item metric employed to quantify physical activity demonstrates documented construct validity in large-scale behavioral tracking, it assesses behavior solely within a temporal-frequency framework. The lack of detailed empirical parameters regarding duration, metabolic intensity, and specific exercise modalities—which are essential for clinical exercise prescription—precludes this model from isolating distinct adaptive physiological thresholds. Future structural models should combine self-reported frequency classifications with multi-axial accelerometry to balance this variance distribution.

Conclusion

Through an empirical investigation of 2144 adolescents, this study identified the protective mechanisms and pathways through which PA alleviates insomnia in adolescents. The main conclusions are as follows:

PA exerts a significant negative predictive effect on insomnia. Regular exercise serves as a robust behavioral buffer directly reduces risks of DIS, DMS and EMA, facilitating stable sleep architecture. Anxiety and SVA act as critical mediators within the exercise-sleep axis. Specifically, PA influences insomnia indirectly through 3 distinct pathways: “PA $\rightarrow$ Anxiety $\rightarrow$ insomnia”, “PA $\rightarrow$ SVA $\rightarrow$ Insomnia”, “PA $\rightarrow$ Anxiety $\rightarrow$ SVA $\rightarrow$ Insomnia”. Exercise habituation serves a dual role of psychological buffering and behavioral reconstruction. Rather than merely serving as a physiological trigger for sleep, PA fosters psychological resilience and lowers affective distress. This optimization of the psychological profile reduces adolescents’ reliance on compulsive media consumption as an avoidant coping mechanism, curtails late-night bedtime procrastination, and ultimately restores sleep homeostasis.

This study confirms the multidimensional efficacy of PA in the prevention and treatment of insomnia, clarifying the intrinsic association of “exercise-psychology-behavior-sleep”. The education department should establish systematic PA programs, emphasize the mental health benefits of exercise, and develop an integrated model for adolescent health promotion that coordinates physiological,

psychological and behavioral dimensions, thereby realize comprehensive guarantee of sleep health among young people.

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